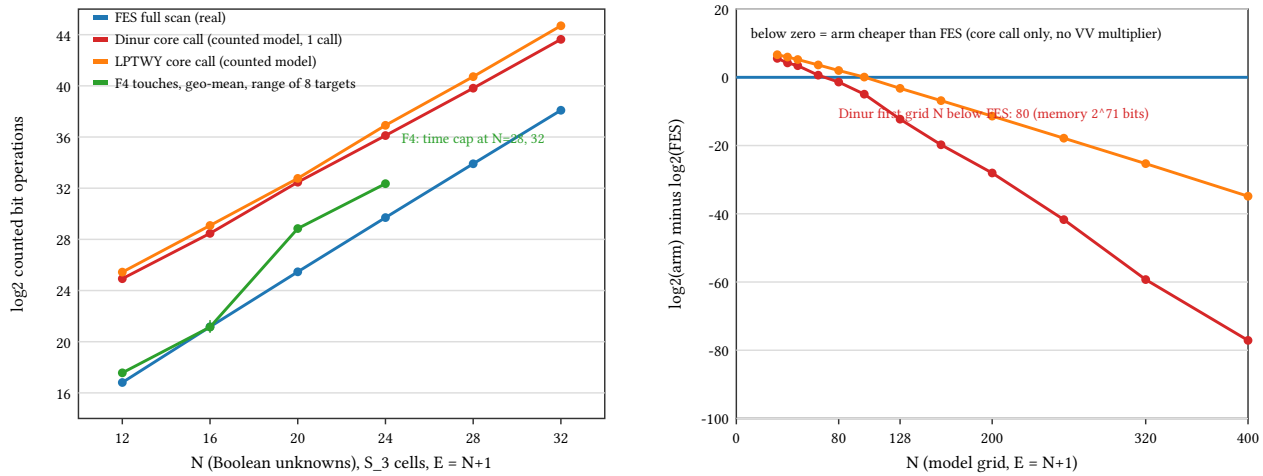
Item 03: polynomial-method solving on descended Semaev systems

Reassessment after the repair pass. Instrument sha256 `c4a2a1ac...`; receipts `raw/receipt-card-01.json`, `raw/receipt-card-02-r2.json` to `10-r2`; summary `raw/summary-r2.txt`.

Verdict

INCONCLUSIVE by the frozen rule (level: none reached). The frozen sensitivity control needs the Dinur count to move by less than 1% across E = 17, 33, 64; measured 1.36% and 3.99%, so it fails. Descriptive branch with that control waived: **TRADES** (exponent over constant). Both arms cost far more than FES at N = 32 (46.8x and 97.7x); their fitted exponent is lower. No improvement over the index-calculus baseline is measured or claimed. msolve arm BLOCKED.

Item 03: polynomial-method cost against FES and F4 on descended Semaev systems (verdict INCONCLUSIVE by the frozen rule)



Source: raw/summary-r2.txt (b03 summarize, source sha256 c4a2a1ac...), raw/card-0N-r2-*.stdout.log. Costs are deterministic; no sampling interval. Host: shared 4-vCPU, isolation L0.
Left: S_3 cells N = 12..32 (measured rows for FES and F4; counted-model rows validated against the real counter on 8856 of 8856 calls). Right: crossover model to N = 400.

Figure 1: Counted cost against N (left, S_3 cells, measured FES and F4, modelled Dinur and LPTWY) and crossover model to N = 400 (right). Source `raw/summary-r2.txt`. Costs are deterministic; no sampling interval. Host shared 4-vCPU, L0.

Numbers (log2 bit operations)

cell (N)	FES real	dinur	lptwy	rho dinur	rho lptwy	F4 touches
c13-6-2 (12)	16.807	24.928	25.440	3.59e-3	2.52e-3	17.570
c17-8-2 (16)	21.170	28.473	29.084	6.33e-3	4.15e-3	21.147
c21-10-2 (20)	25.459	32.474	32.764	7.74e-3	6.32e-3	28.847
c25-12-2 (24)	29.700	36.110	36.909	1.18e-2	6.76e-3	32.357
c29-14-2 (28)	33.907	39.818	40.715	1.66e-2	8.93e-3	cap 80 s
c33-16-2 (32)	38.087	43.637	44.698	2.14e-2	1.02e-2	cap 90 s

Fitted exponent c (jackknife = leave-one-cell-out stability range): FES 1.0632 [1.0571, 1.0683]; dinur 0.9372 [0.9329, 0.9418]; lptwy 0.9666 [0.9594, 0.9795]; F4 1.3016 [1.2564, 1.4097] over four cells.

Crossover (counted model, E = N+1): Dinur below FES from N = 80 on the grid, LPTWY from N = 128; Dinur working set 2^71 bits at N = 80; no crossover under a 2^43-bit cap up to N = 400.

Exponent statement (level: exponent)

Asymptotic c: 0.6946 (d = 2), 0.8115 (d = 3). Decomposition beats rho only if c < 1/2 - 1/m (0 for m = 2, 1/6 for m = 3). Distance remaining 0.6946 and 0.6448; index-calculus exponent relative to rho 2/m + 2c = 2.389 and 2.290 (above 1 loses). Fitted S_3 exponents give 2.87 (dinur), 2.93 (lptwy), 3.13 (FES). No cross-method row exists, so the 0 / 0.1 / 1 GAE pricing band is not applicable.

Controls

Selftest OK. Positive control 1: 8856 real calls, 710 odd (400 required), 0 wrong; real counter equals the model on all. Positive control 2: published exponents reproduced. Negative control: identical model costs on descended and random systems. Sensitivity: failed (above).

Caveats

Shared 4-vCPU host, isolation L0; toy scale N <= 32; F4 hit time caps at N = 28 and 32; costs are lower bounds (one core call, no Valiant-Vazirani multiplier). Revision-card relays were not verbatim; disk logs, which hash-match the receipts, were read.